d. Monounsaturated fat

The correct answer is: Monounsaturated fat

39. While counseling a patient about healthy snacking, a nurse uses a packaged roasted peanut snack as an example. The nutrition label indicates that per 100 g, the product provides 567 kilocalories. The net weight of the entire package is 175 g. If the patient eats the whole package, how many total kilocalories will they consume?
    a. 1,250 kcal
    b. 1,120 kcal
    c. 993 kcal
    d. 850 kcal

The correct answer is: 993 kcal

40. You are a nurse working in a tertiary hospital in Accra, and you are assigned to care for Mr. Smith, a 54-year-old American expatriate who has been admitted with newly diagnosed stage 1 hypertension (BP 148/94 mmHg). During your health education session, he mentions that he regularly shops at international supermarkets in Accra and has access to various food products. He expresses interest in trying dietary modifications before starting medication. You decide to recommend the DASH diet, which was proven in the Dietary Approaches to Stop Hypertension clinical trial to significantly reduce blood pressure. What are the key components of the DASH diet?
    a. A typical American diet with moderate sodium restriction and portion control
    b. A plant-based vegetarian diet with complete elimination of all animal products
    c. A diet rich in fruits and vegetables while maintaining usual dairy and fat intake
    d. A diet rich in fruits, vegetables, and low-fat dairy products with reduced saturated and total fat intake

The correct answer is: A diet rich in fruits, vegetables, and low-fat dairy products with reduced saturated and total fat intake

1. Scientists studying vitamin K found that there was not enough evidence to determine the exact average requirement for the nutrient. Instead, they looked at the typical intakes of healthy adults and set a value that is expected to be sufficient for good health. Which Dietary Reference Intake (DRI) does this value represent?
    a. Recommended Dietary Allowance (RDA)
    b. Estimated Average Requirement (EAR)
    c. Adequate Intake (AI)
    d. Tolerable Upper Intake Level (UL)

The correct answer is: Adequate Intake (AI)

2. Which of these are B-complex vitamins: I. Cobalamin, II. Ascorbic acid, III. Biotin, IV. Molybdenum
    a. I and III ONLY

b. I, II and IV only
c. I, II and III only
d. All of the above

The correct answer is: I and III ONLY

3. Which nutrient can interfere with iron absorption in the body when it is in excess supply?
    a. Biotin
    b. Carbohydrates.
    c. Niacin
    d. Calcium

The correct answer is: Calcium

4. A physician prescribes 1000 mg of calcium supplement daily for a patient at risk of bone loss. As the nurse preparing the medication, what would be the most appropriate course of action?
    a. Administer the supplement as prescribed, since the dosage aligns with the body's requirement for a macromineral like calcium.
    b. Recognize that calcium is a trace mineral and consult the physician about the unusually high dose before giving it.
    c. Withhold the supplement and request a dietary assessment before administration.
    d. Substitute the calcium with a multivitamin that contains smaller quantities of minerals to avoid over-supplementation.

The correct answer is: Administer the supplement as prescribed, since the dosage aligns with the body's requirement for a macromineral like calcium.

5. Kwaku picked up a carton of fortified plant-based milk from the supermarket and placed it in his shopping bag alongside other groceries. When he arrived home, he noticed that the nutrition label had been partially damaged, with the milligram amount of calcium torn away. However, he could still read that the product provides 15% of the Guideline Daily Amount for calcium per 100ml serving, and the footnote clearly stated that the manufacturer used 1000 mg as the Guideline Daily Amount for calcium. What amount of calcium in milligrams was most likely printed in the damaged area of the label per 100ml?
    a. 100 mg
    b. 120 mg
    c. 150 mg
    d. 200 mg

The correct answer is: 150 mg

6. A 39-year-old woman follows a habitual diet consisting mainly of refined carbohydrates, sugar-sweetened beverages, pastries made with hydrogenated oils, and very little intake of fruits, vegetables, or unsaturated fats. Laboratory analysis shows that her blood cholesterol is elevated, and she has particularly low levels of the lipoprotein responsible for transporting cholesterol away from tissues and back to the liver. Which lipoprotein's function has most likely been inhibited by her dietary pattern?
    a. LDL
    b. HDL
    c. VLDL
    d. Chylomicrons

The correct answer is: HDL

7. In nutrition, energy in food is commonly expressed in kilocalories (kcal). Which of the following best defines one kilocalorie?
    a. The amount of energy required to raise the temperature of 1 gram of water by 1°C
    b. The amount of energy required to raise the temperature of 1 kilogram of water by 1°C
    c. The total energy the body uses in one hour of rest
    d. The amount of energy stored in 1 gram of carbohydrate

The correct answer is: The amount of energy required to raise the temperature of 1 kilogram of water by 1°C

8. Individuals experiencing bloating, flatulence, and diarrhea after consuming dairy products are most likely to benefit from which dietary adjustment?
    a. Increasing intake of complex carbohydrates
    b. Reducing consumption of lactose-containing foods
    c. Avoiding foods rich in sucrose
    d. Increasing dietary fiber to enhance digestion of milk sugar

The correct answer is: Reducing consumption of lactose-containing foods

9. A 45-year-old man habitually consumes a diet rich in fatty cuts of red meat, butter, cream, and processed snacks prepared with shortening. Over time, laboratory tests reveal that his blood cholesterol levels have steadily increased. Which lipoprotein receptor is most likely inhibited by the effect of this dietary pattern, leading to the rise in blood cholesterol?
    a. HDL receptor
    b. LDL receptor
    c. VLDL receptor
    d. Chylomicron remnant receptor

The correct answer is: LDL receptor

10. Only a small portion of starch is digested in the mouth by salivary amylase – roughly about 5%. From a nutritional and health perspective, why is this limited breakdown beneficial?
    a. It prevents the mouth from becoming too acidic during digestion
    b. It reduces the amount of glucose available to oral bacteria that cause tooth decay
    c. It ensures that starch digestion starts early for faster absorption
    d. It allows more starch to be fermented by bacteria in the large intestine

The correct answer is: It reduces the amount of glucose available to oral bacteria that cause tooth decay

11. An investigator studying long-standing dietary patterns across southern Europe reports a habitual pattern marked by: predominant use of a monounsaturated culinary oil, plentiful seasonal fruits and vegetables, cereals (mostly unrefined) as staple carbohydrates, moderate fish and dairy intake, and relatively low consumption of red and processed meats. Which named dietary pattern does this description most closely represent?
    a. Traditional Mediterranean dietary pattern
    b. Western (industrialized) dietary pattern
    c. DASH (Dietary Approaches to Stop Hypertension)
    d. Traditional East Asian dietary pattern

The correct answer is: Traditional Mediterranean dietary pattern

12. Which pair of monosaccharides differs most notably in the number of atoms forming their ring structures, where one contains a six-membered ring and serves as the body's main energy source, while the other forms a five-membered ring and is less stable metabolically?
    a. Glucose and Fructose
    b. Galactose and Glucose
    c. Ribose and Deoxyribose
    d. Maltose and Lactose

The correct answer is: Glucose and Fructose

13. A nurse is educating a patient about incorporating healthy fats into their diet. Which statement correctly describes monounsaturated and polyunsaturated fatty acids?
    a. Monounsaturated fats have one double bond (e.g., oleic acid in olive oil and avocado), while polyunsaturated fats have multiple double bonds (e.g., fish oils and canola oil). Both are beneficial for health and are liquid at room temperature.
    b. Monounsaturated fats have multiple double bonds and are found in fish oils, while polyunsaturated fats have one double bond and are found in olive oil. Only monounsaturated fats are beneficial for cardiovascular health.

c. Monounsaturated fats have no double bonds and are solid at room temperature, while polyunsaturated fats have one double bond and are liquid. Only polyunsaturated fats are recommended for heart health.
d. Monounsaturated and polyunsaturated fats both have one double bond but differ in their food sources. Monounsaturated fats should be avoided while polyunsaturated fats are beneficial.

The correct answer is: Monounsaturated fats have one double bond (e.g., oleic acid in olive oil and avocado), while polyunsaturated fats have multiple double bonds (e.g., fish oils and canola oil). Both are beneficial for health and are liquid at room temperature.

14. A nurse is counseling a client who reports frequent consumption of fried fast foods, baked snacks, and vegetable oils commonly used in processed foods. The client rarely eats nuts, seeds, green leafy vegetables, or fish. Which of the following nutritional imbalances is the client most likely to have, and what is the main concern associated with it?
    a. Low omega-6 intake leading to poor absorption of fat-soluble vitamins
    b. Excess omega-6 intake relative to omega-3 intake, increasing risk of inflammatory and chronic diseases
    c. High omega-3 intake relative to omega-6 intake, leading to bleeding and bruising tendencies
    d. Balanced omega-3 and omega-6 intake because both come from plant oils

The correct answer is: Excess omega-6 intake relative to omega-3 intake, increasing risk of inflammatory and chronic diseases

15. The digestibility of carbohydrate-rich foods depends in part on the molecular arrangement of their storage polysaccharides. One form consists of long, mostly unbranched chains of glucose, while another form has extensive branching, creating more accessible points for enzymatic action. In humans, the storage polysaccharide is even more highly branched than its plant counterpart, allowing for rapid energy mobilization when needed. Which of the following correctly identifies these three molecules in order of least branched to most branched?
    a. Glycogen → Amylopectin → Amylose
    b. Amylose → Amylopectin → Glycogen
    c. Amylopectin → Amylose → Glycogen
    d. Amylose → Glycogen → Amylopectin

The correct answer is: Amylose → Amylopectin → Glycogen

16. For potassium, the Dietary Reference Intake provides an Adequate Intake (AI) of 4,700 mg/day for adults. Based on this information, which of the following statements is most correct?
    a. An Estimated Average Requirement (EAR) and Recommended Dietary Allowance (RDA) have also been established for potassium.
    b. There is no EAR, and therefore no RDA, for potassium.

c. The AI is always higher than the RDA for any nutrient.
d. The AI was established because potassium has a Tolerable Upper Intake Level (UL).

The correct answer is: There is no EAR, and therefore no RDA, for potassium.

17. According to the Acceptable Macronutrient Distribution Range (AMDR) guideline, what is the recommended average range of total protein intake per day for an adult male consuming a 2,500 kcal diet?
    a. 40 – 120 g
    b. 50 – 150 g
    c. 63 – 219 g
    d. 75 – 250 g

The correct answer is: 63 – 219 g

18. In nutritional assessment, certain dietary methods are vulnerable to reactivity (i.e. subjects altering their usual intake because they know their diet is being monitored). Among the following methods—(I) 24-hour recall, (II) estimated food record, (III) weighed food record, (IV) food frequency questionnaire—which one(s) is/are most clearly subject to reactivity bias in practice?
    a. I only
    b. II and III only
    c. I, II and III
    d. II, III and IV

The correct answer is: II and III only

19. When foods rich in starch and fibre, such as grains or tubers, are chewed for a long time, they begin to taste slightly sweet. What best explains this observation?
    a. Fibre molecules are converted directly into glucose during chewing
    b. Salivary amylase breaks down starch into maltose, which tastes sweet
    c. Chewing releases natural fructose trapped in plant cells
    d. Bacteria in the mouth ferment starch to produce simple sugars

The correct answer is: Salivary amylase breaks down starch into maltose, which tastes sweet

20. Nurse Kwame is teaching a diabetes education class about the essential nutrients the body needs to function properly. He explains that all foods contain combinations of six classes of nutrients that serve different roles in the body. Which option correctly lists all six classes of nutrients?
    a. Fruits, vegetables, grains, protein, dairy, and fats
    b. Carbohydrates, proteins, fats, vitamins, minerals, and fiber
    c. Carbohydrates, proteins, fats, vitamins, minerals, and water
    d. Sugars, starches, proteins, fats, vitamins, and water

The correct answer is: Carbohydrates, proteins, fats, vitamins, minerals, and water

21. Ama is counseling a patient about sodium intake and references a canned tomato sauce nutrition label. The label indicates that one serving (1/2 cup) contains 480 mg of sodium, which represents 20% of the Daily Value. The patient asks the dietitian what total daily sodium intake this percentage is based on. To provide an accurate response, what Reference Daily Intake (RDI) value for sodium should Ama calculate that the manufacturer used?
    a. 1,920 mg
    b. 2,400 mg
    c. 2,000 mg
    d. 2,300 mg

The correct answer is: 2,400 mg

22. Carbohydrates exist in several forms, including starch, glycogen, cellulose, and simple sugars such as glucose, fructose, and galactose. Which of these is known to taste the sweetest to humans?
    a. Glucose
    b. Fructose
    c. Sucrose
    d. Galactose

The correct answer is: Fructose

23. In human nutrition, minerals are categorized based on the quantities required by the body. A patient under nursing care is prescribed a supplement that provides less than 100 mg each of zinc, copper, and selenium daily. From a nutritional standpoint, how should these minerals be classified, and what does this classification imply about their physiological importance compared with minerals such as calcium and phosphorus?
    a. They are ultratrace minerals, meaning they have no known physiological function in humans.
    b. They are major electrolytes, implying they primarily function to maintain fluid balance and nerve transmission.
    c. They are trace minerals, indicating they are needed in minute amounts but play critical roles in metabolic and enzymatic functions.
    d. They are macrominerals, meaning they are required in large amounts for structural functions like bone formation.

The correct answer is: They are trace minerals, indicating they are needed in minute amounts but play critical roles in metabolic and enzymatic functions.

24. The UL for vitamin D in adults is 100 µg (4000 IU) per day. What does this value indicate?
    a. It is the recommended daily intake for maintaining bone health.
    b. Intakes above 100 µg per day are always harmless.
    c. Consuming vitamin D in amounts higher than 100 µg per day may pose health risks.

d. There is no established risk even if vitamin D intake exceeds this value.

The correct answer is: Consuming vitamin D in amounts higher than 100 µg per day may pose health risks.

25. Which of the following foods is considered whole grain?
    a. White bread
    b. Banku made from wet-milled whole maize dough
    c. Gari (cassava flakes)
    d. White rice (polished)

The correct answer is: Banku made from wet-milled whole maize dough

26. Maria is a 28-year-old marketing professional whose estimated energy requirement (EER) is 2357 kcal per day. Following AMDR guidelines, she aims to get 55% of her daily calories from carbohydrates to fuel her active lifestyle and demanding work schedule. How many grams of carbohydrates should Maria consume daily to meet this target?
    a. 289 grams
    b. 324 grams
    c. 361 grams
    d. 1296 grams

The correct answer is: 324 grams

27. After food leaves the mouth and enters the stomach, what happens to the digestion of carbohydrates?
    a. It continues rapidly because stomach acid activates amylase
    b. It slows down because salivary amylase is inactive in the acidic environment
    c. It stops completely because peristaltic movements are absent in the stomach
    d. It increases because the stomach secretes enzymes that digest starch

The correct answer is: It slows down because salivary amylase is inactive in the acidic environment

28. Which of the following statements accurately describes the structural and health characteristics of cis and trans fatty acids?
    a. In cis fatty acids, hydrogen atoms are bonded to opposite sides of the carbon chain creating a straight structure, while trans fatty acids have hydrogen atoms on the same side creating a bent structure. Trans fats are naturally occurring in foods and have been associated with reduced risk of coronary heart disease according to the Harvard Nurses Health Study.
    b. In cis fatty acids, hydrogen atoms are bonded to the same side of the carbon chain creating a bent structure, while trans fatty acids have hydrogen atoms on opposite sides creating a straighter structure. Trans fats

are primarily produced through industrial hydrogenation, though small amounts occur naturally in ruminant animals. The Harvard Nurses Health Study linked trans fatty acids with increased risk of coronary heart disease due to their negative impact on blood cholesterol levels.

c. Both cis and trans fatty acids have hydrogen atoms bonded to the same side of the carbon chain, but trans fats undergo hydrogenation which makes them solid at room temperature. Trans fats from all sources, including those naturally found in dairy products, have been associated with coronary heart disease according to the Harvard Nurses Health Study.
d. In trans fatty acids, hydrogen atoms are bonded to the same side of the carbon chain, while cis fatty acids have them on opposite sides. Trans fats are naturally found in tropical oils and vegetable oils. The Harvard Nurses Health Study found no significant association between trans fatty acids and coronary heart disease risk.

The correct answer is; B

29. You are a nurse working in a tertiary hospital in Accra, and you are assigned to care for Mr. Smith, a 54-year-old American expatriate who has been admitted with newly diagnosed stage 1 hypertension (BP 148/94 mmHg). During your health education session, he mentions that he regularly shops at international supermarkets in Accra and has access to various food products. He expresses interest in trying dietary modifications before starting medication. You decide to recommend the DASH diet, which was proven in the Dietary Approaches to Stop Hypertension clinical trial to significantly reduce blood pressure. What are the key components of the DASH diet?
    a. A typical American diet with moderate sodium restriction and portion control
    b. A diet rich in fruits and vegetables while maintaining usual dairy and fat intake
    c. A diet rich in fruits, vegetables, and low-fat dairy products with reduced saturated and total fat intake
    d. A plant-based vegetarian diet with complete elimination of all animal products

The correct answer is: A diet rich in fruits, vegetables, and low-fat dairy products with reduced saturated and total fat intake

30. In nutritional assessment, one method involves (a) conducting a 24-hour recall of actual foods and beverages consumed, (b) eliciting information about the individual's usual eating pattern (meal timing, habits, changes over time), and (c) administering a food frequency questionnaire to verify and clarify the earlier data, while also asking for usual portion sizes (often in household measures) so that nutrient intakes can be calculated via food composition tables. This integrated method is particularly useful when one wants to assess an individual's habitual diet in a clinical or research setting. Which of the following methods is being described?

a. 24-hour recall
b. Estimated food record
c. Food frequency questionnaire
d. Dietary history

The correct answer is: Dietary history

31. Disaccharides in the human diet are formed when two monosaccharides join through glycosidic bonds. Glucose, a central energy molecule, is present in all three nutritionally important disaccharides, paired with different partners. This explains why glucose intake rises whenever sucrose, maltose, or lactose is consumed. Which of the following correctly matches glucose with its disaccharide partner?
    a. Glucose + Glucose → Sucrose
    b. Glucose + Galactose → Maltose
    c. Glucose + Galactose → Sucrose
    d. Glucose + Galactose → Lactose

The correct answer is: Glucose + Galactose → Lactose

32. A packaged serving of instant oatmeal with dried fruits is sold in a 100 g packet. The manufacturer's label shows that 100 g provides 60 g of carbohydrates, 12 g of protein, and 8 g of fat. It also supplies Vitamin B1 (1.1 mg), Iron (10 mg), and Calcium (400 mg). A student consumes only 75 g of the oatmeal. Based on this portion size, calculate the total energy value obtained.
    a. ≈270 kcal (Calories)
    b. ≈340 kcal (Calories)
    c. ≈360 kcal (Calories)
    d. ≈380 kcal (Calories)

The correct answer is: ≈270 kcal (Calories)

33. The dietary reference intake of a certain nutrient is set at 2,500 mg/day. In the study population from which this value was derived, half of the individuals had nutrient consumption above this level. Which Dietary Reference Intake (DRI) does this most likely describe?
    a. Acceptable Macronutrient Distribution Range (AMDR) for Linolenic acid
    b. Tolerable Upper Intake Level (UL)
    c. Estimated Average Requirement (EAR)
    d. None of the above

The correct answer is: Estimated Average Requirement (EAR)

34. When a person consumes large amounts of carbohydrate-rich foods such as rice, bread, and sugary drinks, and the glycogen stores in the liver and muscles are already full, which of the following best describes what happens to the excess glucose in the body?